

An Improved CSP Algorithm and Application in Motor Imagery Recognition¹

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Abstract—An improved common special pattern (CSP) algorithm was proposed to improve classification accuracy of motor imagery electroencephalograph (EEG) in a brain-computer interface (BCI) system. Three channels EEGs were filtered through the time window and band-pass filter to obtain the most obvious features of event-related desynchronization and event-related synchronization. Improved CSP algorithm combined with support vector machine (SVM) as adopted for the classification of motor imagery EEGs. The experiment results show that the improved CSP algorithm can avoid the repetitive eigenvector selection and discriminate the left hand and right hand mental task more accurately than the traditional CSP.

Keywords—brain-computer; interface; CSP; SVM; recognition;

INTRODUCTION

There are some kinds of diseases which can damage the neuromuscular communication between brain and the outside environment, such as amyotrophic lateral sclerosis (ALS). These diseases cause people to lose their movement ability gradually and at a later stage people may become totally paralyzed. Although modern life-support technology can prolong patients' life, they live with low quality of life and bring their families and society heavy burden. People try to build a kind of communication and control channel directly between the brain and external devices, without the participation of peripheral nerves and muscles, which named Brain Computer Interface (BCI) [1].

BCI technology has evoked an increasing interest between many international and domestic researchers. It shows an important research value and application prospect among different fields, such as rehabilitation medicine, auxiliary control, entertainment, military, and so on. Although there is a rapid improvement of BCI research qualitatively and quantitatively, it is still in the experimental stage because of two major problems: low recognition rate and slow communication velocity [2-5].

Effective feature extraction method is an important approach to improve the recognition accuracy. There are several feature extraction methods, such as: (1) Power Spectral Analysis. Power Spectrum Analysis can reflect variances of signals' power, but the time information may be lost. (2) Adaptive Autoregressive (AAR). AAR model can reflect variances of event-related electroencephalograph (EEG) without prior information of related bands and is suitable for online analysis. However, artifacts should be controlled effectively in application of this method. (3) Wavelet Transform. Zooming characteristics of wavelet transform can amplify the discrimination between different classifications. That can help to improve recognition accuracy. At present, the spontaneous EEG, based on prior knowledge, extract wavelet coefficients of interest frequency band as features. Nevertheless, because of the complexity of EEGs mechanism, it is difficult to obtain accurate prior knowledge. (4) Common Spatial Pattern (CSP). The goal of CSP is to design spatial filters which lead to new signal matrix whose variances are optimal for the discrimination of two classes of EEGs. Specific frequency bands need not to be pre-selected, but multi-electrode are required for the EEG analysis which limits the application of CSP in portable BCI system. So it is necessary to optimize the number of electrodes further [6-9].

Improving the signal processing speed is another prerequisite of realizing BCI signals on-line processing. It is determined by two main factors: the amount of data to be processed and the complexity of the algorithm. Two countermeasures are put forward to the above two factors: reducing the electrodes number and proposing an effective algorithm.

In this paper, CSP algorithm is applied to the feature extraction in three EEG electrodes. Furthermore, a new feature extraction definition is proposed to overcome the repeated selection of eigenvectors, combined with Support Vector Machine (SVM) for the classification of motor imagery EEGs.

□. EXPERIMENTAL DATA

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A Data source

EEG data of motor imagery provided by GRAZ BCI center is adopted in this paper. It is divided into two parts: training set and testing set. Each set includes 140 trials: 70 left trials and 70 right trials.

This dataset was recorded from a normal subject during a feedback session. The experiment consists of 7 runs with 40 trials each. All runs were conducted on the same day with several minutes break in between. The subject sat in a relaxing chair with armrests. Three bipolar EEG channels (anterior '+', posterior '-') were measured over C3, Cz and C4. The EEG was sampled with 128 Hz. The task was to control a feedback bar by means of imagery left or right hand movements. Each trial continues 9s. The first 2s was quite, at $t=2s$ an acoustic stimulus indicates the beginning of the trial. The order of left and right cues was random[13]. Electrode positions are shown in figure 1:

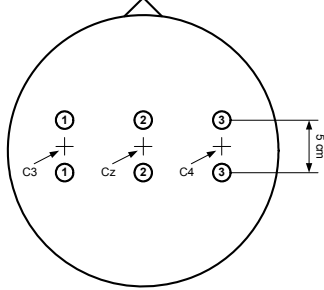


Figure 1: Electrode positions (left) and timing scheme (right).

B Data preprocessing

Based on the physiological phenomenon ERD (Event-Related Desynchronizations) and ERS (Event-Related Synchronization), EEGs provided by GRAZ university were filtered through a 8-30 Hz frequency band-pass filter. Let TC_3 and TC_4 be the average instantaneous power of C_3 and C_4 of N trials respectively.

$$TC_3 = \frac{1}{N} \sum_{i=1}^N C_3^2 \quad (1)$$

$$TC_4 = \frac{1}{N} \sum_{i=1}^N C_4^2 \quad (2)$$

Let LTC_3 and LTC_4 denote the average N trials instantaneous power of C_3 and C_4 for the left hand motor imagery task. RTC_3 and RTC_4 are the same for the right hand. LTC_3 , LTC_4 and RTC_3 , RTC_4 are shown in figure 2 and figure 3 with $N=70$.

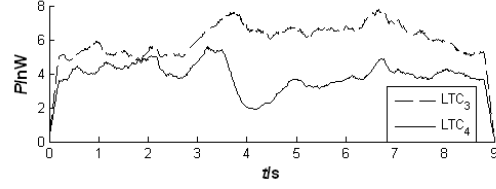


Figure.2 Average power of C_3 and C_4 during left hand imaging movement

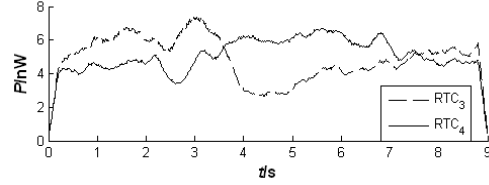


Figure.3 Average power of C_3 and C_4 during right hand imaging movement

From Figures 2 and 3, obvious variance of power of C_3 and C_4 confirms the ERD / ERS physiological phenomenon and provides a theoretical basis for feature extraction.

C. Feature Extraction Based on Improved CSP Algorithm

The CSP method is based on a decomposition of the raw EEG signals into spatial patterns, which are extracted from two populations of single trial EEG. These patterns maximize the difference between the populations.

In data processing of multi-channel EEGs, two or more best eigenvectors are selected which can describe the even-related EEG, and do variance operation with them to define the feature vector. However, when this method is applied to process three-channel EEGs, it will lead to repeated selection of eigenvectors and decrease the classification accuracy.

In order to avoid this problem, traditional CSP algorithm is modified in this paper. The best eigenvector corresponding to the biggest eigenvalue was selected so that the state of motor imagery EEGs is described best and the repeated selection of eigenvalues is avoided effectively. Then recognition accuracy is improved. The detailed steps are as follows.

From figure 2, the difference between C_3 and C_4 at 4 ~ 5s interval is obvious, so 40 samples are selected in this period for feature extraction. In condition that $XL \in R^{3 \times 40}$ and $XR \in R^{3 \times 40}$ are computed respectively by averaging over all trials to express those 40 samples of left-hand and right-hand motor imagery. Then the covariance matrix CL and CR can be obtained as following.

$$CL = [XL] \times [XL]^T \quad (3)$$

$$CR=[XR] \times [XR]^T \quad (4)$$

The composite covariance matrix C and its eigenvalue decomposition is given by

$$C=[CL] + [CR]=F_c \phi F_c^T \quad (5)$$

where $F_c \in R^{3 \times 3}$ is a matrix of normalized eigenvectors corresponding to matrix $\phi \in R^{3 \times 3}$ of eigenvalues. The whitening transformation equalizes the variances in the space spanned by the eigenvectors in F_c .

$$P = \phi^{-\frac{1}{2}} F_c^T \quad (6)$$

CL and CR will be transformed to $\overline{CL}$ and $\overline{CR}$ using whitening matrix P as the follows:

$$\overline{CL} = P \times CL \times P^T = U \lambda U^T \quad (7)$$

$$\overline{CR} = P \times CR \times P^T = U(I - \lambda)U^T \quad (8)$$

Where U is the eigenvector matrix corresponding with diagonal matrices λ or $(I-\lambda)$. Then spatial filter can be constructed as following:

$$W_L = U_L^T \times P \quad (9)$$

$$W_R = U_R^T \times P \quad (10)$$

Where $W_L \in R^{1 \times 3}$ corresponds to the left hand, and $W_R \in R^{1 \times 3}$ to the right hand. Then signal matrices XL and XR are filtered by the filters W_L and W_R respectively.

$$Z_L = W_L \times XL \quad (11)$$

$$Z_R = W_R \times XR \quad (12)$$

So, $Z_L \in R^{3 \times 1}$ and $Z_R \in R^{3 \times 1}$ will maximize the differences of two classes.

Based on CSP algorithm, the feature vector f is defined as (13) in multi-electrode EEG collecting.

$$f = \frac{VAR(Z_L)}{VAR(Z_L) + VAR(Z_R)} \quad (13)$$

When this method is adopted, the feature vector f for three-channel EEG will be scalar, and the event-related EEG cannot be described accurately. Thus it will lead to lower classification accuracy. In this paper, a new feature vector is defined as:

$$f = [Z_L, Z_R] \quad (14)$$

Where $f \in R^{1 \times 80}$, including the EEG variance of 40 samples after the occurrence of the movements completely, can reflect the state of EEGs better.

D. Recognition algorithm based on Support Vector Machine

As a state-of-the-art classification methodology, SVMs have sound theoretical foundations and provide excellent performance in various pattern recognition applications. The basic idea of SVMs is to look for the optimal decision hyperplane which best separates the data points into two classes with a maximum margin in a projected feature space based on the Structure Risk Minimization principle. With respect to the recognition of small sample, nonlinear and high-dimensional data, SVM performs better adaptability, stronger classification ability and higher computational efficiency. Therefore, SVM is selected for EEG recognition of motor imagery in this paper [10-12].

With fewer parameters and lower complexity, Radial Basis kernel function (RBF) is commonly used as SVM kernel function. Because of nonlinear properties of EEGs, RBF is selected as (15) for nonlinear mapping from low-dimensional space to high-dimensional space.

$$k(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right), \gamma > 0 \quad (15)$$

Where x_i is the i -th input, γ is the kernel parameter

Kernel parameter γ and error punishment factor C are main parameters which affect the performance of SVMs. γ decides the distribution of transformed data in space, and punishment factor C determines the convergence speed and generalization ability of SVMs. Therefore, γ and C are closely related to the recognition rate.

Grid search method is adopted for the determination of parameter γ and C . In other words, the value ranges of γ and C are made as the grid edges, and different values of γ and C cross over different grids. The optimal value γ and C are found by searching on all of these grids. In particular, coarse grid and fine-mesh are combined to reduce the computational complexity and accelerate the search speed.

γ and C vary as 2^n . For coarse grid, the step of n is 1, and for fine-mesh the step of n is 0.1. By using the method for GRAZ dataset, the result is shown as figure 4. The recognition accuracy of left hand imagery movement can get to 92.86%, and the recognition accuracy of right hand imagery movement EEGs attains 91.43%.

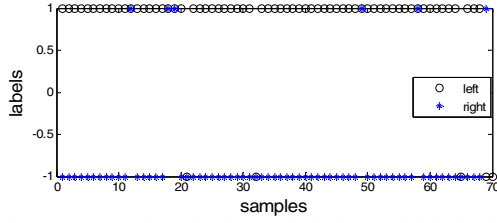


Figure.4 EEGs classification result for left and right hand imaging movement

In order to illustrate the advantages of the feature definition proposed in this paper, the dataset is deal with traditional CSP algorithm too. The results as shown in table 1, and it shows that the new feature definition avoids the repeated selection of eigenvectors and has a better classification performance.

TABLE I COMPARISION OF CLASSIFICATION RESULTS

CSP+SVM	Accuracy (left)	Accuracy (right)	Accuracy (average)
New feature definition	92.86%	91.43%	92.145%
Traditional feature definition	75.7%	87.14%	81.425%

III. CONCLUSION

CSP spatial filtering method can maximize the difference in variances of two kind of motor imagery EEGs. Moreover, the new feature definition in this paper, compared with traditional CSP feature definition, can improve the recognition rate in condition of three-channel EEG and develop a new idea for portable BCI system.

However, when more EEG information are used to distinguish different mental tasks, it will lead to high dimension of input vectors and make against generalization of the algorithm. Therefore, the later work will focus on how to reduce the dimension of input vector and realize the analysis online.

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